Baldivis Secondary College

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Year 9 **End of Term** 3 Test

/50______________Name:_________________

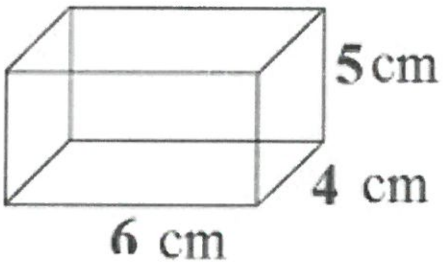
(One page of Notes, Formula sheet & Calculators allowed)

If ALL WORKING is not shown, FULL MARKS may not be given.

<u>CALCULATOR Section:</u>

Volume

1. Find the volume of this rectangular prism

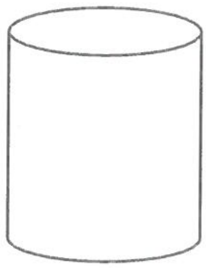


$5 \times 4 \times 6 = 120\ cm^3$ ✓ ✓

(2 marks)

2. Find the volume of this cylinder

H: 10m



Radius: 5m

$\pi \times 5^2 \times 10$ ✓

$785.4\ m^3$ ✓

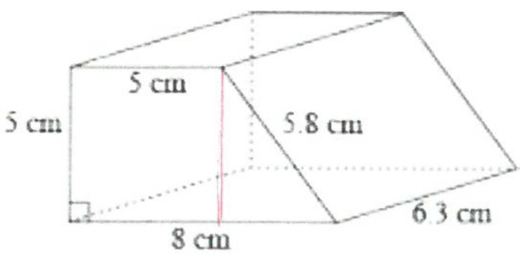
or

250π

(2 marks)

3. Find the volume of the following composite shapes

a)



$\left(\frac{5+8}{2}\right) \times 5 \times 6.3 =$ ✓

6.5

$204.75\ cm^3$ ✓

(3 marks)

or

$5 \times 5 \times 6.3 = 157.5\ cm^3$

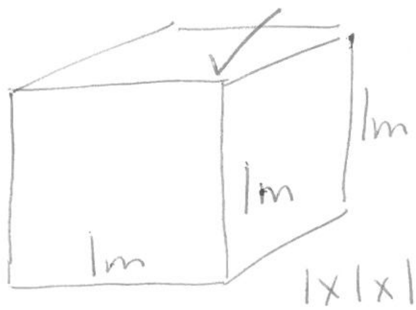
$+\ 47.25$

$204.75\ cm^3$

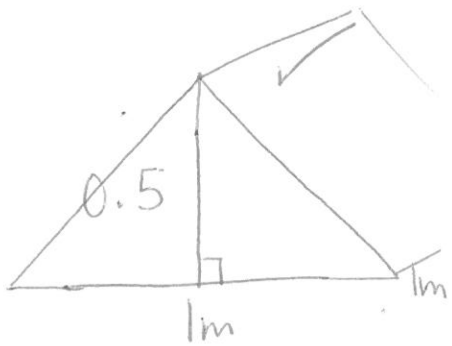
4) Find the volume of this dog house

1m x 1m x 1m
with the roof peak 0.5m higher

i) What two prisms do you need to find the volume of the kennel? Draw a diagram of each.



$1 \times 1 \times 1 = 1m^3$



$1 \times 1 \times 0.5 = 0.5m^3$

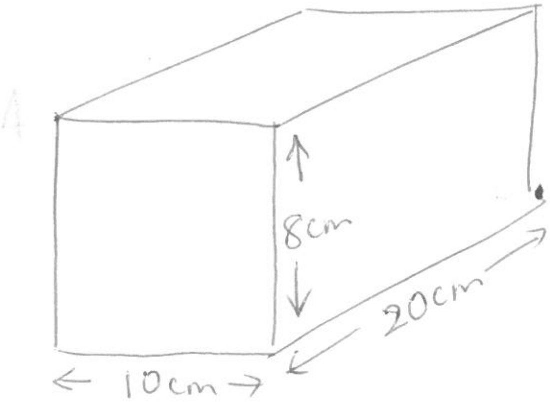
ii) Put dimensions on the diagram

iii) Calculate the volume of the dog house (7 Marks)

$1 + 0.5 = 1.5m^3$

5) A fish tank measures 20cm x 10cm x 8 cm

How much water will I need (L) to fill the tank?



$20 \times 10 \times 8 =$
$1600 cm^3 = 1600 mL$
$= 1.6 L$

(3 marks)

6.

The open surface of a pond has an area of 46.6 m2. The pond is known to be 86cm deep and the area of the base is the same as the area of the surface. If the pond is half mud, what is the volume of mud in this pond?

$\frac{86}{2}$

$46.6 m^2 \times$
43
$= 2003.8 m^3$

(2 marks)

g) If a 13 cm high cylinder holds 375mls, how wide is its radius and diameter rounded to the nearest cm?

(4 marks)

Financial Maths: Simple Interest

1. Daniel wants to buy a car so he borrows $8 000 from **Just Do It Loans** Company

   He agrees to pay 15% per annum (pa) interest over a 5-year term.

   Find the Simple Interest for his loan
   $8000 x 17/100 x 5 = $
   How much needs to be re-payed in total?

(2 Marks)

2. Calculate the Simple Interest to be paid on
   a) $700 invested for 1 year at 10% int. pa

   b) $500 invested for 8 years @ 6 % pa

   c) $5 000 invested for 3 years at 4.95 % pa

(3 marks)

$1600.

3. If Mr Magoo deposits his tax refund into his bank account at simple interest earning a rate of 3 % pa, for 4 years,
   a) How much interest would he earn from the bank?
   b) How much total investment would he have at the end of 4 years? (Hint: TR)

$1600 \times 0.03 \times 4 = \$192$

(~~4~~ 2 2 marks)

$$\begin{array}{r} \$192 + \\ 1600 \\ \hline \underline{\underline{\$1792}} \end{array}$$

4. An investment earns $620 as simple interest at 8% pa over 4 years. What was the Original amount (Principal) invested?

$\$620 = 0.08 \times 4 \times P$

$P = \dfrac{I}{R \times T}$

$P = \dfrac{620}{0.08 \times 4} = \underline{\underline{\$1937.50}}$

(3 marks)

wording.

5. The ~~yearly~~ amount of..... interest that appeared in Lu Jane's bank account was $6,754.34 after an invest of 7 years. every year.

What interest rate was being paid if the Original amount invested was $261,532?

(Hint; Make R the subject of the formula)

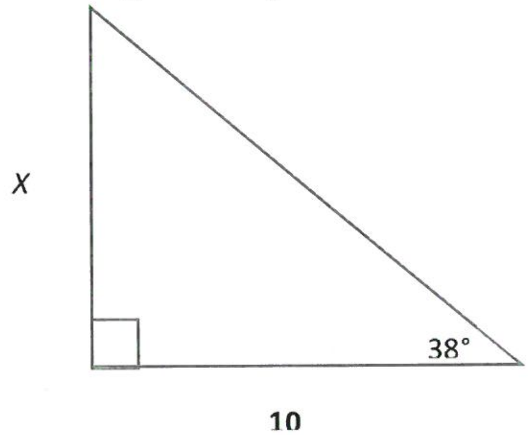
$R = \dfrac{I}{P \times T} = \dfrac{6754.34}{7 \times \$261532}$

(5 marks

$R = 0.0037 = 0.37\%$

Trigonometry

1. Use the Tan ratio to find the missing side length.

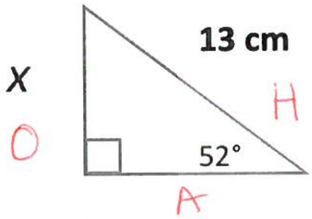


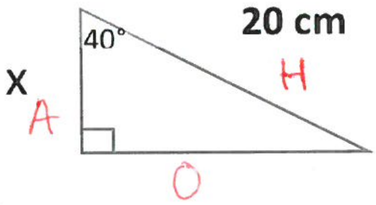
Tan 38° x 10 = x ✓

7.81 ✓

(2 mark)

2. a) Label the sides





b) determine the Trig ratios

Soh Sin (as O x)

Cos (H as x) Cah. Cos $\frac{x}{20}$.

c) Calculate the missing side lengths (1 dp.)

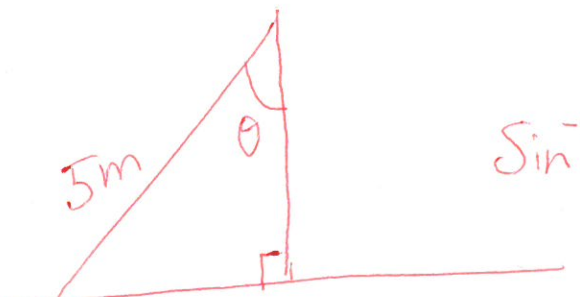
10.2

Cos x 20 = x.

= 15.32 cm

(8 Marks)

3. A 5m ladder is against a wall. If the base is 2.5m from the wall, what angle does the ladder make with the wall?



2.5m

$\sin^{-1}\left(\frac{O}{h}\right) = \sin^{-1}\left(\frac{2.5}{5}\right) = 30°$.

(4 marks)

END OF TEST

23
ST = 15
Trig = 14.
29
51